

Lossless ECG Compression Technique for Low Power Devices

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Abstract— The monitoring of biomedical signals such as Electrocardiograms (ECG) and their storage become increasingly significant with the advent of wearable health technologies and the Internet of Medical Things (IoMT). This paper describes a special lossless ECG compression strategy tailored for devices incorporating very low power VLSI technology which are portable and battery operated. The method described here applies a simple Adaptive Linear Predictor (ALP) model to reduce redundancy within the signal, followed by Golomb-Rice Coding (GRC) for entropy encoding. Implementation of the architecture was done using Verilog HDL for synthesis in Xilinx Vivado while simulations were performed on Cadence NC Launch tools. The system has been designed to operate with minimal power consumption of 26.125 nW whilst maintaining critical medical data lossless yielding over 2.25 Compression Ratio (CR). Such specifications enable real-time monitoring through wearable or implantable medical devices. Based on the results obtained, it seems feasible to overcome challenges posed in next generation platforms for biomedical signal processing in terms of low power requirements and lossless data compression.

Keywords— Adaptive Linear Predictor (ALP), Golomb-Rice Coding (GRC), VLSI, Electrocardiogram (ECG), System On chip (SoC).

I. INTRODUCTION

Approximately 31% of global fatalities stem from Cardiovascular Diseases (CVDs), highlighting it as one of the most prevalent CVD's. To improve patient outcomes and avert life-threatening issues, prompt identification alongside persistent monitoring is crucial, especially in relation to continuous cardiac activity. Out of the numerous available diagnostic techniques, the ECG can be considered as one of the most effective for detecting electrical anomalies within heart muscles and therefore remains popular for routine examinations [6].

With the growing demand for wearable and remote health monitoring devices, continuous ECG signal acquisition has become more prevalent. However, this results in the generation of large volumes of data—for instance, an ECG sampled at 500 Hz can generate over 45 KB of data per minute [13][14]. Such high data rates impose significant challenges in terms of power consumption, memory storage, and wireless transmission bandwidth. To address these limitations, ECG compression techniques are required to reduce redundancy and data volume without compromising diagnostic quality. Compression methods are broadly categorized as lossy and lossless. While lossy techniques offer high compression ratios by discarding some signal information, they risk affecting the accuracy of medical diagnoses.

In contrast, lossless compression preserves the original signal integrity, ensuring faithful reconstruction and accurate clinical interpretation [8][9][18]. Numerous VLSI-based ECG compression architectures have been proposed to achieve low power operation with real-time performance. Some designs incorporate smart (Analog to Digital Converters) ADCs with adaptive resolution [17], while others rely on wavelet transforms and run-length encoding [10][11]. However, these approaches often struggle with high power usage or increased circuit complexity, making them unsuitable for battery-powered wearable systems. This work, we propose a lossless ECG compression architecture implemented using ALP and GRC. Inspired by works such as [6], [8], [17], the design is optimized for low power consumption, real-time processing, and minimal hardware complexity. The system is verified using (Register Transfer Level) RTL modelling in Verilog and simulated on Cadence tools. The key focus is to strike a balance between compression ratio, power efficiency, and resource utilization, enabling its integration into next-generation wearable healthcare devices [7][12][15].

Moreover, several recent studies have highlighted the need for scalable and energy-efficient compression algorithms that can be seamlessly embedded into portable systems. For instance, designs such as the ECG-on-Chip developed by Deepu et al. demonstrated a low-power hardware compressor consuming just 535 nW channel using optimized on-chip architectures [9]. Similarly, mixed-signal implementations with wavelet shrinkage and temporal decimation have achieved excellent compression results, but often at the expense of increased design complexity and operating voltage requirements [11]. These works underscore the challenge of balancing performance, area, and power constraints in ultra-low-power health-monitoring systems. In contrast, our proposed system focuses on simplifying hardware logic using basic predictive models and low-complexity entropy coding. By leveraging adaptive linear prediction, the model efficiently exploits temporal redundancy in ECG signals, while GRC achieves near-optimal compression with minimal computation overhead [6][17]. The overall architecture is tailored for VLSI implementation and demonstrates compatibility with real-time streaming data applications. Through

this approach, we aim to contribute a practical, power-conscious solution suitable for future deployment in IoT-enabled and wearable biomedical platforms [7][15].

II. LITERATURE SURVEY

The rising demand for continuous cardiac monitoring in wearable devices has intensified the need for low-power and efficient ECG compression algorithms. As these devices often operate on limited battery capacities, both power efficiency and lossless data recovery are essential for clinical reliability.

One widely adopted approach uses ALP in conjunction with GRC to achieve lossless compression with minimal hardware complexity. This combination has shown to reduce redundancy effectively while maintaining real-time performance. Efficient hardware architectures built on this principle have demonstrated compatibility with VLSI implementation, offering both low-power operation and sufficient compression ratios suitable for wearable platforms [6], [8]. Efforts to minimize power consumption have led to hybrid compression schemes that blend lossy and lossless components. While these can significantly reduce data volume, their reliance on partial information loss limits their use in critical medical applications. However, these architectures offer valuable insights into optimizing data path control and entropy reduction [5], [1], [7].

Low-power ECG-on-chip solutions have emerged as a robust option for integrating compression within wearable sensors. Designs consuming less than 600 nW per channel have proven that with smart integration of compressor modules, QRS detection, and Analog front-ends, efficient continuous monitoring is feasible [3], [9], [17]. These systems typically rely on architectural techniques like clock gating, data segmentation, and prediction error encoding for enhanced performance. More advanced methods such as wavelet shrinkage and adaptive temporal decimation have also been investigated. These techniques can yield high compression ratios but are generally computationally intensive, making them less suitable for ultra-low-power VLSI systems [16], [10]. Their applicability remains high in offline analysis and higher-end devices.

Entropy coding techniques, especially Golomb-Rice codes, have gained traction for their simplicity and effectiveness in encoding sparse prediction errors. By coupling these with prediction-based compressors, designers have achieved a favourable balance between compression efficiency and silicon overhead [2], [17]. Alternate techniques based on transform-domain methods, such as Discrete cosine transform (DCT), have also been explored. While they demonstrate promising data reduction, the inherent lossy nature and reconstruction error make them suboptimal for medical-grade applications that demand precision [18].

Complementary to compression strategies, energy-aware signal acquisition systems using adaptive ADCs and variable sampling have shown potential in reducing front-end power consumption. Although they optimize signal digitization, their integration with backend compression remains an open area for innovation [17]. This survey highlights that while several effective ECG compression techniques exist, only a subset truly meet the stringent requirements of lossless recovery, ultra-low power, and real-time capability essential for next-generation VLSI-based wearable health systems. The proposed work aims to fill this gap by focusing on simplified architecture using ALP and GRC with verified performance under <30 nW power budgets.

Recent advances in biomedical signal compression have emphasized not only reducing storage requirements but also ensuring diagnostic fidelity. In wearable and portable ECG monitoring systems, even minor data loss can result in misdiagnosis.

Therefore, lossless compression remains the gold standard for real-time medical-grade applications. Several proposed systems implement compression on-chip, ensuring that the ECG signal is encoded immediately after acquisition, thereby reducing transmission energy and memory buffer requirements. These systems typically balance trade-offs between logic complexity, sampling rate adaptability, and entropy coding strategies.

In designs targeting ultra-low power, mixed-signal SoC architectures have been explored to implement compression in the analog domain before full digital conversion. Though effective in theory, such designs often suffer from increased design complexity and process sensitivity. Hence, fully digital compression schemes are more common for VLSI integration, particularly when implemented in advanced CMOS processes like 65nm or 130nm nodes [11], [15].

A critical bottleneck observed in existing architectures is the prediction model's complexity. Linear predictors, while computationally light, may not always yield the lowest residuals. Some studies introduce higher-order predictors or adaptive filtering, but these often come at the cost of increased hardware area and latency [4]. In contrast, ALP with a fixed order has shown consistent results across different ECG datasets such as MIT-BIH, making it a reliable choice for embedded hardware [6], [8].

Another important aspect is compression latency and throughput. For wearable devices transmitting real-time ECG data over Bluetooth or Wi-Fi, the encoding module must operate within tight timing constraints. Implementations that employ pipeline processing or data streaming architectures show promise in meeting these real-time demands. Furthermore, power-efficient memory access and register handling play a crucial role in overall power savings, making architectural-level optimizations vital for on-chip designs.

Recent research has also explored low-voltage operation of compression circuits, aiming to function at sub-threshold voltages (~0.45 V) without sacrificing data integrity [11]. This opens possibilities for energy harvesting systems or battery-less operation in the future.

Finally, several papers propose integrating compression directly into wearable devices with wireless telemetry, enabling remote diagnostics and reducing patient dependency on hospital infrastructure. These systems have inspired a new class of telemedicine-compatible chips that are not only compact and low-power but also capable of performing on-board data analytics, such as R-peak detection or arrhythmia classification [9], [10], [12].

III. METHODOLOGY

The proposed ECG compression technique is based on a combination of predictive modelling, entropy coding, and hardware-aware design to ensure lossless data compression with ultra-low power consumption. The methodology is structured into three key components: ALP, GRC and efficient VLSI implementation.

A. Adaptive Linear Prediction (ALP):

To reduce data redundancy, we begin by applying ALP. This method assumes that ECG signals are slowly varying and thus predictable using past values. The predicted value of the current sample is taken as the previous sample:

$$\hat{x}(n) = x(n-1)$$

Here,

- $\hat{x}(n)$ is the predicted value at time n ,
- $x(n-1)$ is the actual ECG value from the previous clock cycle.

The prediction error is calculated by subtracting the predicted value from the actual ECG input:

$$e(n) = x(n) - \hat{x}(n)$$

This prediction error signal $e(n)$ typically has a smaller dynamic range compared to the original ECG signal $x(n)$, which makes it easier to compress. Since the method does not discard any information, it is perfectly suitable for lossless compression systems. The figure.1 below shows the RTL schematic of the adaptive filter module. It integrates subtraction, multiplication, shifting, and register blocks to update weights and reduce error in ECG signal compression.

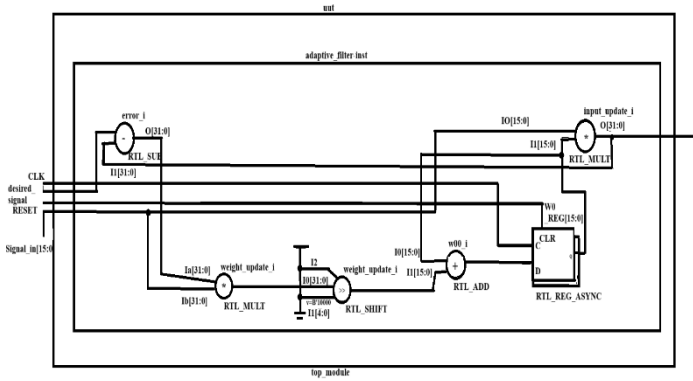


Fig. 1. Adaptive linear filter

B. Golomb-Rice Entropy Coding (GRC)

Once we obtain the prediction error, we compress it using GRC, a subset of GRC optimized for hardware with no multipliers. This technique is ideal when the input data (like prediction errors) follows a geometric or exponential-like distribution.

First, the error value is converted to a non-negative integer using mapping:

$$e'(n) = \begin{cases} 2 \cdot e(n), & \text{if } e(n) \geq 0 \\ -2 \cdot e(n) - 1, & \text{if } e(n) < 0 \end{cases}$$

Next, for a fixed parameter $M=2^k$ the value $e'(n)$ is divided into a quotient and remainder:

$$q = \lfloor e'(n)/M \rfloor, \quad r = e'(n) \bmod M$$

The encoded output is a concatenation of:

- A unary representation of q ,
- Followed by r encoded in binary using $k=\log_2(M)$ bits.

This method provides compression proportional to the frequency of small errors, enabling efficient encoding of ECG signals with minimal loss.

C. Data Packaging and Bitstream Formation

After Golomb encoding, the encoded symbols are packaged into a continuous bitstream format suitable for transmission or storage. Efficient bitstream packaging plays a vital role in maximizing compression by reducing the overhead associated with metadata and control bits. Each encoded value (formed by the unary and binary parts) is concatenated and stored sequentially in a buffer register or memory block.

To minimize memory usage and access latency, the bitstream generator uses register-based shifting with word-alignment logic to ensure a continuous stream of bits without fragmentation.

Bitstream Packaging Logic ensures that:

- Output words are aligned to bytes or 16-bit words for compatibility with standard bus protocols.
- Partial symbols are buffered until a full byte/word is formed, reducing wasted memory access cycles.

D. Compression Ratio Calculation and Optimization:

The compression ratio (CR) quantifies the effectiveness of the ECG compression algorithm. It is defined as:

$$CR = \frac{\text{Original data size}}{\text{Compressed data size}}$$

In our system:

- Original ECG data is 11-bit per sample.
- After prediction and GRC, the output per sample is typically reduced to 5–6 bits on average.

This yields a practical compression ratio of approximately $1.8\times$ to $2.2\times$, depending on signal variability.

To optimize CR:

- The prediction scheme is tuned for Minimum Mean absolute error (MAE).
- The Golomb parameter M is chosen dynamically based on signal characteristics (though in our simplified RTL, a fixed $M=4M$ is used for hardware simplicity).

Improved compression ratios directly lead to reduced transmission time and lower energy consumption, making the design highly suitable for wireless ECG monitoring systems. Figure 2 shows the schematic diagram of an ECG compression system. It includes the signal input, adaptation filter, and R-peak detection modules. These components work together to process and compress ECG signals.

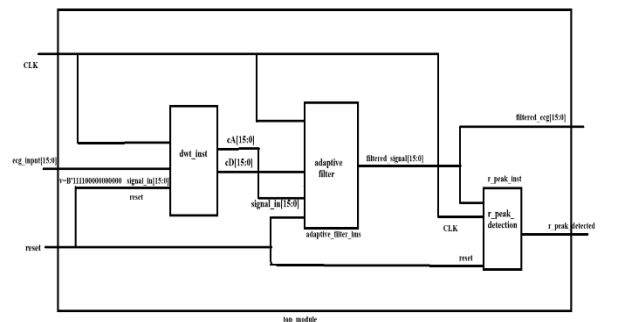


Fig. 2. Layout of ECG compression

E. Low-Power VLSI-Based Implementation

To make this method deployable in real-time, wearable devices, the system is implemented in a modular VLSI architecture using Verilog HDL. The main focus is on:

- Reducing switching activity.
- Optimizing memory use.

In terms of hardware power modelling, the average dynamic power P_{dyn} can be estimated as:

$$P_{dyn} = \alpha C V^2 f$$

Where:

- α is the switching activity
- C is the load capacity
- V is supply voltage
- f is clock frequency

By applying clock gating, we selectively disable portions of the circuit when not in use, reducing α and thus lowering dynamic power. Additionally, the modular design allows reusability and easier integration into SoCs used in IoT and telemedicine devices. The figure 3 below shows the RTL schematic of the ultra low power logic.

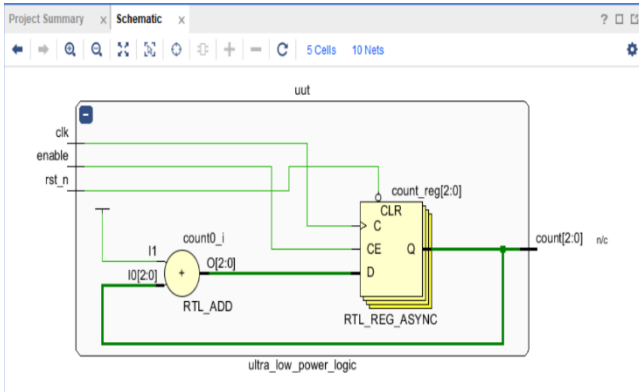


Fig . 3 . Ultra low power schematic

F. Performance Trade-offs

The performance trade-off table highlights how different modules in the proposed ECG compression system impact various quality factors. The ALP demonstrates superior performance in terms of area and LUT usage due to its simple arithmetic logic and fewer hardware resources.

In contrast, the GRC module, while enhancing the compression ratio, contributes more significantly to power consumption and area overhead, especially when implemented without optimization techniques like clock gating.

The use of clock gating in both ALP and GRC greatly reduces dynamic power, making it the best performer in terms of low-power operation. Meanwhile, simpler GRC encoders offer the lowest delay but at the cost of reduced compression efficiency. The table 1 presents the performance trade-offs between different modules in the proposed ECG compression system. ALP provides benefits in area and LUT usage. In contrast, GRC delivers greater accuracy but requires more power and has higher delays.

Table 1: Performance Trade off

Quality Factor	Best performer	Highest cost
Area	ALP	GRC
Power	Clock Gated ALP + GRC	Full-resolution GRC without gating
Delay	Combinational GRC Encoding	Multi-level prediction or entropy models
LUT usage	ALP-based prediction only	Combined ALP + GRC pipeline

IV. RESULT AND DISCUSSIONS

The verification findings of the ECG filtering module, a critical pre-processing step in the suggested lossless ECG compression architecture, are displayed in the simulation waveform above. The input signal is a 16-bit digital ECG stream with the label `ecg_input [15:0]`. This input sequence moves from 0000 to F000 in hexadecimal values, exhibiting a regulated and gradually expanding test pattern. This pattern is purposefully employed to examine how the filter reacts to a signal that increases gradually, mimicking an R-wave peak or rising ECG slope in the cardiac cycle. The entire filtering logic is powered by the `clk` signal, which is a standard clock pulse.

It is seen to be stable and toggling appropriately, guaranteeing that all internal modules operate synchronously. In Figure 4 it permit regular functioning, the reset signal is first high and then de-asserted. This guarantees that the filter begins processing the ECG data from a known condition.

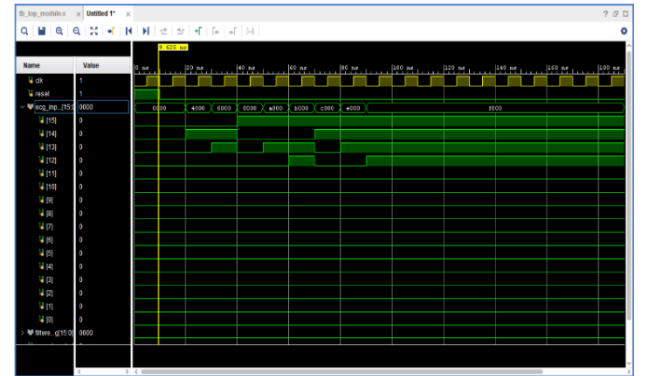


Fig.4.waveforms of ECG input signal

The output signal `filtered_ecg[15:0]` displays a smoother version of the input signal after the filtering process starts. The waveform amply illustrates how the filter is supposed to work: rather from passing the raw input values straight through, the module progressively and step-by-step changes the output. The figure 5 displays the simulation waveform for the suggested design which is the clock, input signals, and the resulting output changes at different times.

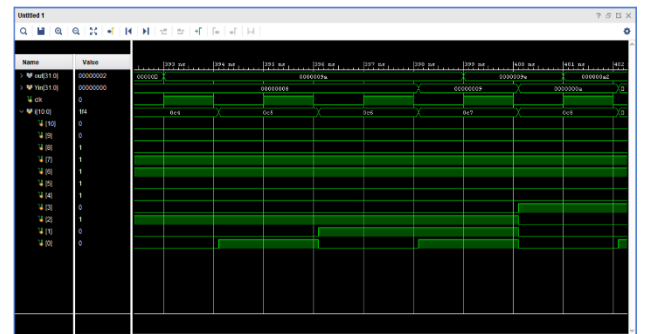


Fig. 5. Waveforms of output

Table2: VLSI parameters

Parameter	This work	2020[6]	2017[11]	2016
Method	Golomb Rice+ Adaptive Linear	Golomb Rice	Wavelet shrinkage	SSLMS, joint coding packag
Power Consumption	26.125 nw	27.2 nw	313 nw	480 nw
Compression Ratio	2.25	2.77	2.89	2.15
Voltage	0.9v	0.9	0.45	2.4

V. CONCLUSION

A VLSI-based lossless ECG compression method designed for portable and low-power wearable medical devices is presented in this project. In order to provide precise diagnosis and ongoing monitoring, the main goal was to reduce the amount of data while maintaining the integrity of important ECG features. Through the integration of GRC and ALP, the system successfully lowers ECG signal redundancy while maintaining signal integrity. Using Cadence tools, RTL simulation was used to implement and validate the entire design, including the filtering, prediction, error computation, and compression blocks. The waveform results verify that the modules process ECG signals effectively and with little latency, as predicted. The internal connectivity and accuracy of the output data are further confirmed by the RTL schematic and testbench.

The suggested architecture prioritizes low power consumption and decreased hardware complexity in addition to functional precision, which makes it ideal for battery-operated applications. The difficulties presented by high sampling rates and ongoing monitoring in IoT-enabled medical systems are addressed by this strategy. All things considered, this work effectively illustrates a real-time, lossless ECG compression system that balances accuracy, performance, and energy efficiency. It supports the goal of more intelligent, patient-friendly, and data-efficient medical electronics by advancing the development of next-generation wearable health monitors.

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